CODEFORCES
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F. An Ancient Sorting Algorithm

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

You are given an array of N integers. You have to sort the array in non-decreasing order using a custom sorting algorithm with the following constraint:

- You may only swap adjacent elements with the same parity (i.e., both even or both odd)

Sort the array in non-decreasing order until no more such swaps are possible and print the final array.

Input

The first line contains a single integer N $(1 \leq N \leq 1000)$ — the number of elements in the array.

The second line contains N integers separated by spaces $a_1, a_2, a_3 \ldots a_n$ $(1 \leq a_i \leq 10^6)$ — the elements of the array.

Output

Print the final array after sorting the array in non-decreasing order until no more such swaps are possible.

Examples

input	Copy
7 4 2 4 7 1 6 1	
output	**Copy**
2 4 4 1 7 6 1	

input	Copy
5 3 5 9 7 1	
output	**Copy**
1 3 5 7 9	

input	Copy
14 4 8 2 9 1 5 4 6 8 1 7 13 11 8	
output	**Copy**
2 4 8 1 5 9 4 6 8 1 7 11 13 8	

input	Copy
1 221	
output	**Copy**

CSE221 Assignment 01 Fall 2026
Contest is running
9 days
Virtual Participation

→ Submit?
Language: Java 21 64bit
Choose file: Choose File No file chosen
Success!
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